CLASS 9 · CBSE · GANITA MANJARI PART 1 · DPP SHEET

Chapter 2: Introduction to Linear Polynomials

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Write the degree of $7 - 4x^3 + x$ and of the constant polynomial 12.
2. In $5x^3 - x^2 + 4$, write the coefficient of x^2 and the constant term.
3. Find the value of $3x - 7$ at $x = -2$.
4. Find the value of $2t^2 - 5t + 1$ at $t = 3$.
5. One number is 9 more than another, and their sum is 71. Find the numbers.
6. Find the rule for the nth term of the pattern 4, 7, 10, 13, … and its 20th term.
7. A tank holds 500 litres and drains at 35 litres an hour. Write the volume left after t hours and find it for $t = 8$. Is this growth or decay?
8. An electricity bill is ₹260 for 8 units and ₹380 for 14 units. If the bill is $y = ax + b$ for x units, find a and b.
9. Write the slope and the y-intercept of $3y = 9x + 6$.
10. Which of these lines are parallel: $y = 4x + 1$, $y = -4x + 1$, $2y = 8x - 5$?

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Answer Key

1. 3 and 0
2. -1 and 4
3. -13
4. $18 - 15 + 1 = 4$
5. 31 and 40
6. $3n + 1$; 61
7. $V = 500 - 35t$; 220 litres; linear decay
8. $a = 20, b = 100$
9. $y = 3x + 2$: slope 3, y-intercept 2
10. $y = 4x + 1$ and $2y = 8x - 5$ (both have slope 4)

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